



Proposal: AIOps Agent for Root Cause Analysis with Business Impact

Proposal: AIOps Agent for Root Cause Analysis with Business Impact

Problem Statement 3:

AIOps Agent for Root Cause Analysis with Business Impact

Proposed Idea & Approach (Key Parameters)

I propose an AI-powered AIOps agent that automatically analyzes each IT incident using technical data (logs, metrics, incident metadata) and business data (customer profile, SLA, revenue/process criticality).

For every incident, the agent will generate:

- Root cause summary with short reasoning (2–4 bullets)
- Remediation suggestion in safe step-by-step format
- Business impact message in plain language, including estimated financial loss per hour and priority level (P1/P2).

Key parameters considered:

- Error frequency and log patterns
- Latency, CPU, memory, and availability anomalies
- Incident duration and affected service/component
- Customer transaction volume, industry criticality, revenue impact, SLA penalty
- Model/Techniques to be Used
- Data preprocessing and feature extraction from logs and metrics
- Machine learning classification model (e.g., XGBoost/Random Forest) for root-cause prediction
- Rule-based runbook mapping for remediation steps
- Business impact calculation model using customer profile data:
- Loss per hour = Revenue impact + SLA penalty + operational disruption cost
- NLP/LLM-based summarization to generate clear human-readable explanation and impact message.

High-Level Steps / Methodology

- Merge incident technical data and customer business profile data into one incident view.
- Clean and transform logs/metrics; detect anomalies and extract important features.
- Train and apply a classification model to predict likely root cause category.
- Generate short explanation of diagnosis based on top signals from logs/metrics.

- Recommend remediation actions from predefined runbook steps linked to root cause type.
- Estimate business impact and financial loss per hour using business parameters.
- Assign priority (P1/P2) using impact level and criticality rules.
- Produce final incident output for support teams:
- Root cause + reasoning
- Fix steps
- Business impact statement + €/hour + severity

Expected Outcome

The proposed solution will help support teams identify incidents faster, take safer corrective action, and understand customer/business impact immediately. This improves response quality, prioritization, and overall operational efficiency.